

Writing a Literature Review in Engineering

How to use this guide

Literature reviews can be tricky; that's why we have this guide! This guide is structured to step you through the process of developing a literature review.

Important! Always check your assessment task sheet and marking rubric for specific requirements for your literature review. This guide helps you with the process, but your final submission must comply with your assessment task requirements.

What is a literature review?

A literature review is a critical summary of published sources (the ‘literature’) on a particular topic. Doing a literature review helps you answer three core questions for your research project:

- What do we already know about the topic (i.e., previous research findings)?
- How reliable is this knowledge (i.e., quality of previous research)?
- What are the research ‘gaps’ that your project can contribute to?

Literature reviews also develop the skills required to meet Engineers Australia’s competency standards for Professional Engineers:

Bachelor-level students need to meet the [Stage 1 Competencies](#) to become professional engineers:

- 1.4.a) **Identifies and critically appraises** current developments, advanced technologies, emerging issues and interdisciplinary linkages in at least one specialist practice domain of the engineering discipline.
- 3.2.a) **Is proficient in** listening, speaking, reading and writing English, including: comprehending critically and fairly the viewpoints of others [. . .]

Masters-level students need to meet the [Stage 2 Competencies](#) to become professional engineers:

- 8. **Deliver clear written and oral presentations** on engineering problems and engineering activities in English [. . .]
- 12. **Develop and apply current research papers** to inform and shape perceptions of engineering possibilities to meet [client] needs
- 14. When you identify or are presented with engineering problems, **adopt appropriate research methods** to locate previously known solutions to similar problems [. . .]

Key characteristics of literature reviews

- Some sources have a section called ‘literature review’ but not all. It’s common for the literature review to be included in the ‘introduction’ or ‘background’ sections of engineering journal articles.
- Referencing: Literature reviews have lots of in-text citations. This is because literature reviews need to talk about the literature in order to review it.
- Summarising: Good literature reviews provide summaries of a field of research. For example, ‘Pedestrian crashes with vehicles have so far been extensively investigated.’ Summarising sources demonstrates that you have read widely and understand the field of research.

- Critiquing: Good literature reviews critique individual sources and fields of research. Critiquing demonstrates that you can identify the strengths and limitations of research.

Activity

Read this [example literature review on pedestrian injury in bus-pedestrian crashes](#).

Consider the extent to which it addresses the three core questions:

- What do we already know about the topic (i.e., previous research findings)?
- How reliable is this knowledge (i.e., quality of previous research)?
- What are the research 'gaps' that your project can contribute to?

Check your understanding

True or false: A literature review is just a summary of existing knowledge on a topic.

Answer: False. While a literature review does summarise existing knowledge, it must be done in a critical way. This means that the writer evaluates the quality of previous research as well.

Getting started

Important! Have you agreed on your research question with your supervisor?

Having a clear research question is necessary for starting your literature review. If you haven't confirmed your research question, do that with your supervisor first then come back to this guide.

Overview of the literature review process

Step 1. Scope literature review

Scoping your literature review is where you set clear parameters for what types of sources will be included and excluded from your literature review.

Step 2. Find sources

Searching for sources involves identifying keywords, developing search strategies, and selecting and saving sources within your scope.

Step 3. Prepare to read

Purposeful reading is efficient reading. This step is where you will identify what types of information you need to extract from sources and set up a note-taking system.

Step 4. Read and take notes

This is the step where you finally get to read your sources! Using your note-taking system as your reading will make your structuring and drafting steps much easier.

Step 5. Plan structure

In this step you will identify themes that run through the sources. You will then organise these themes into a logical order, which becomes your literature review structure.

Step 6. Draft and edit

Time to write! If you have consistently used your note-taking system and planned your literature review structure, you should have a clear sense of what to write. The drafting and editing step will involve several drafts, feedback processes and proofreading.

Activity: Map out a timeline

Writing a literature review requires time management skills. Creating a timeline will help you manage your literature review process, reduce your stress and result in higher quality work.

Take 5 minutes right now to **map out a timeline**. Consider:

- When do you need to have completed your literature review?

- How long will you take to complete each of the steps?
- What competing demands will you need to juggle at the same time?

Check your understanding

True or false: A literature review can be done quickly and mostly involves writing.

Answer: False. Writing the literature review is the last step in the process. You will spend considerable time reading, note-taking, identifying and organising themes before you start drafting.

Step 1: Scope your literature review

Important! Have you agreed on your research question with your supervisor?

You need a clear research question before you can scope your literature review. If you haven't confirmed your research question, do that with your supervisor first then come back to this guide.

What is 'scope'?

'Scope' refers to the extent of the subject matter that your literature review will address.

The scope of your literature review will be determined by:

- Your research question
- Contextual factors (in your case, your assessment requirements)

Example

The aim of this research was to examine the crash injury severity of pedestrians involved in road traffic crashes in South-East Queensland. The notes below are the student's first attempt at brainstorming the scope of their literature review.

Keywords:

- In scope: crash, incident, collision, accident, "injury severity", fatality, pedestrian
- Out of scope: "car crash", "motor vehicle collision"

Context:

- In scope: urban, city, metropolitan
- Out of scope: regional

Date of publication:

- In scope: last 10 years

Source types:

- In scope: academic journal articles, government publications, professional publications
- Out of scope: industry videos/podcasts, The Conversation articles

Activity: Define the scope of your literature review

Make notes based on the following prompts. You may wish to use [QUT's template](#).

Based on your research question:

- Keywords: What are the keywords in your research question?
- Context: Which contexts are relevant to your research project?

- Date of publication: At what point in time does literature lose relevance to your project?
- Location: Does the location of research matter for your project?

Based on your assessment brief and marking rubric:

- Source types: Which types of sources are required?
- Number of sources: How many sources do you need to include?
- Date of publication: Is there a specified data cut off?
- Are there other scoping factors that are important to your research project?

Check your understanding

True or false: You should read everything that's relevant to your research question.

Answer: False. You should read everything within the scope of your literature review. Your scope will exclude sources on your topic. For example, you may exclude an Engineers Australia report because your task requires scholarly sources.

Step 2: Find sources

Searching for sources involves identifying keywords, developing search strategies, and selecting and saving sources within your scope.

Locating sources

Locating relevant and credible sources requires cyclical approaches to searching. The most effective process for finding sources is the Information Searching Lifecycle:

1. Identify your topic keywords.
2. Build your search strategy.
3. Choose the right tool(s) for what you are wanting to find.
4. Run your search and evaluate your results.
5. Repeat step 4 as necessary, evaluating your results and then refining and adapting your search strategy.

Watch [this video](#) (1:47) for an explanation about how the lifecycle works.

QUT Library Search

The QUT Library homepage is your starting point to finding sources for your review. The Library Search is a powerful tool that can locate a wide range of sources, such as journal articles, conference proceedings, books and book chapters, reports, theses, patents, and even videos!

Watch [this video](#) (1:42) for a quick introduction on how to use the Library Search.

If you've already mastered the basics, check out QUT Library's Study Smart guide for [search techniques](#), [using databases](#), [improving your search strategy](#) and more.

Activity: Try QUT Library's advanced search

1. **Open** your list of keywords. (Go to Step 1 if you haven't scoped your literature review yet).
2. **Review** the keywords you identified as being in scope. Are these still the best keywords for your literature review? Edit if required.
3. **Open** the [QUT Library website](#) and **go to** Advanced search.
4. **Experiment with searching** your keywords and **applying filters** on the search results page. See if you can get your total search results under 100.
5. **Document** your search strategy to avoid duplicating at a later date. You may like to use [QUT's template](#).

If you have previous experience with searching techniques, you may wish to directly search [QUT's engineering databases](#).

Selecting sources

If you have scoped your literature review, selecting sources should be a straightforward task. Use your scoping notes to help you select sources.

Learn more about [evaluating sources](#) with QUT Library's Study Smart guide.

Saving sources

Once you have identified highly relevant sources for your literature review, you should immediately save them. You have a couple of options:

- Save article PDFs into an organised filing system on your device. A consistent naming convention will help you to easily find articles later. For example:
FirstAuthorSurname_Year_Title
- You can use [reference management software](#) (e.g., Mendeley) that is made for this specific purpose. However, using specialist software is not required for your study.

Check your understanding

True or false: Filtering search results is an effective way to locate sources that are in scope for my literature review.

Answer: True. Keyword searching often returns hundreds (if not thousands) of search results. Filtering for a certain date range and/or subject area is an effective way to locate the most relevant sources to your project.

Step 3: Prepare to read

Purposeful reading is efficient reading. This step is where you will identify what types of information you need to extract from sources and set up a note-taking system.

What information do you need to extract from sources?

Let's revisit the purpose of a literature review to answer this question. Doing a literature review helps you answer three core questions for your research project:

- What do we already know about the topic (i.e., previous research findings)?
- How reliable is this knowledge (i.e., quality of previous research)?
- What are the research 'gaps' that your project can contribute to?

Therefore, you need extract information that answers these questions. Information like:

- Research aim
- Findings/conclusions
- Methods used
- Strengths of the research
- Limitations of the research

Setting up a note-taking system

A good note-taking system will increase your reading efficiency and make the structuring and drafting phases much easier. This is because it guides your reading and organises your notes. If you don't have an established note-taking system, try using [QUT's template](#).

Check your understanding

True or false: Setting up a note-taking system before I start reading will reduce re-reading later.

Answer: True. While you may need to revisit some articles, using a note-taking system helps to focus on the most important information for your literature review the first time you read. It also enables you to extract that information and store it in an organised way.

Step 4: Read and take notes

This is the step where you finally get to read your sources! Using your note-taking system as you read will make your structuring and drafting steps much easier.

Read the source and extract information

- Only read one source at a time.
- Start by familiarising yourself with the topic and structure of the source (e.g., skim read the headings). Is this source still in scope for your literature review?
- Use your note-taking system to remind you of what you're looking for in the source. You may like to use QUT's template.
- As you find the information you need, write a dot point version in your note-taking system.

Critique the source

Once you've read the source and extracted the key information for your literature review, it's time for you to think critically about the source as a whole:

- What are the strengths of the research?
- What are the limitations?

Add notes about your critical evaluation to your note-taking system.

Activity: Your ideal reading environment

Active reading and note-taking takes a lot of brain power, so setting up a productive reading environment is essential for your success.

1. **Think** about past experiences where you have done deep reading. **Consider:** What worked? What didn't?
2. **Brainstorm** your ideal reading environment. **Consider:**
 - Where is it?
 - What time of day is it?
 - How long is your reading session?
 - What helps you stay focused?
3. **Make a plan** for how you will achieve this ideal reading environment.

Check your understanding

True or false: I should only take notes on information that's in the source.

Answer: False. You should also take notes on your critical evaluation of the source. What you think the strengths and the limitations of that research are.

Step 5: Plan structure

In this step you will identify themes that run through the sources. You will then organise these themes into a logical order, which becomes your literature review structure.

The basic structure of a literature review is the same as many styles of writing:

- Introduction
- Body paragraphs
- Conclusion

Introduction

One paragraph that addresses:

- What is your topic?
- Why is it important to research?
- What are the aims of your literature review?

Body paragraphs

Each body paragraph will unpack a major idea you have identified in the literature. The total number of body paragraphs will be dependant on what you have identified.

Body paragraph structure

- What is a major idea in the literature?
- What do researchers agree and disagree about on this idea?
- What are the strengths and limitations of the sources that address this idea?

Conclusion

One paragraph that addresses:

- What do we already know about the topic (i.e., previous research findings)?
- How reliable is this knowledge (i.e., quality of previous research)?
- What are the research 'gaps' that your project can contribute to?

How to identify themes in the literature

A theme is anything that is repeated across sources. Themes can be topics, theories, methods or findings. The simplest way to identify themes is to colour code your notes:

1. **Reorganise** your notes so that categories of information are together. Everything on research topics is together, methods together, findings together, strengths together, limitations together and anything else you've been taking notes on. If you have been using QUT's template, this is already done!

2. Look at just one category of notes. **Colour code** anything that's a repeated topic, theory, method or finding. These are themes!
3. **Repeat** Step 2 with each category of information.

Note: It's likely you will identify LOTS of themes (more than you can include in your literature review). Only the most significant themes will become a body paragraph.

Check your understanding

True or false: Body paragraphs should include several references.

Answer: True. Each body paragraph should address one theme. A theme is anything that is repeated across sources. Therefore, your body paragraphs should reference more than one source.

Step 6: Draft and edit

Time to write! If you have consistently used your note-taking system and planned your literature review structure, you should have a clear sense of what to write. The drafting and editing step will involve several drafts, feedback processes and proofreading.

Strategies for writing your first draft

The job of the first draft is to get something on the page. This means your first draft will be ugly. This is how all academic sources start! Here are some strategies that can help you get started:

- Let the first draft be ugly and don't give yourself a hard time about it
- Remove distractions
- Experiment with dictation technologies (e.g., the Dictate feature in Microsoft Word) to say your first draft, rather than writing it down
- Write your first draft in the language you are most comfortable with

Strategies for dealing with procrastination

Procrastination is a normal response to challenging tasks. Here are some strategies that can help you deal with procrastination:

- Try the [Pomodoro Technique](#), which involves timed cycles of work and rest (e.g., 25 minutes working + 5 minutes break)
- Find a writing buddy and help each other stay on task

Strategies for editing and proofreading

Editing is where any piece of academic writing gets its shine. Here are some strategies that can help you edit effectively:

- Plan to have time dedicated to editing.
- Try to wait a day or two before editing so you have 'fresher eyes'.
- Focus on [higher-order concerns first](#) (e.g., checking the structure of your literature review) and lower-order concerns later (e.g., sentence structure).
- Use tools like the [Academic Phrasebank](#) to refine your sentence structure and language.

Strategies to seek feedback

Seeking and applying feedback is a very effective way to improve the quality of your work. Here are some strategies you can use to seek feedback:

- Access [QUT's writing feedback services](#)
- Offer to do a feedback swap with another student, where there is no risk of [collusion or plagiarism](#)
- Make use of generative AI platforms, [if permitted for your assessment](#)

Check your understanding

True or false: I should focus on grammar, spelling and formatting when I start editing my literature review draft.

Answer: False. Higher-order concerns, like checking you've fulfilled the task, are more important and should be prioritised. Editing lower-order concerns, like grammar, should be left till last.

Help available

QUT has Librarians, Educators and Peer Learning Facilitators available for you to ask questions and get feedback.

Researching help

- [Chat online](#) with a Librarian right now.
- Make an on campus or online [appointment](#) with a Librarian.
- Read up on the [Library's searching tips](#).
- Ask your supervisor or unit coordinator about the most relevant publications for your project.
- If you can't find an appointment that suits you, email library@qut.edu.au to request an appointment, providing your preferred dates, times and campus/online.

Writing help

- Develop your academic writing skills at regular [drop-in sessions on-campus or online](#).
- Get feedback on your draft through [Studiosity](#).
- Ask your supervisor or unit coordinator about project requirements and standards.

Referencing help

- [Chat online](#) with a Librarian right now.
- Make an on campus or online [appointment](#) with a Librarian.
- Use the [CiteWrite guide](#).

If you can't find an appointment that suits you, email library@qut.edu.au to request an appointment, providing your preferred dates, times and campus/online.

Academic integrity help

- Complete the [Academic Integrity online module](#).
- Ask your supervisor or unit coordinator about project requirements and standards.

Uni getting complicated?

If you're feeling a bit stuck and would like some support to get back on track, you can [request a call from the Student Success team](#) for assistance to connect with support resources and services.

You can contact the Faculty of Engineering Student Success Team at
success.eng@qut.edu.au